

DSE&DBD

WEEK-3

Name: Thanvi

RollNo:2520030214

1.Document Modelling

```
test> use bookflowDB;
| db.createCollection("book_metadata");
switched to db bookflowDB;
bookflowDB;> db.book_metadata.insertOne({
|   title: "The Silent Algorithm",
|   author: "R. Menon",
|   genre: "Technology",
|   publishedYear: 2022,
|   formats: ["Hardcover", "Digital", "Audiobook"],
|   specs: {
|     pages: 340,
|     fileSize_MB: 4.2,
|     drm: true,
|     narrator: "K. Rao",
|     runtime_min: 610
|   },
|   avgRating: 4.6,
|   reviews: [
|     {
|       member_id: "M1001",
|       rating: 5,
|       comment: "Excellent deep dive into digital systems design.",
|       date: ISODate("2023-01-15")
|     },
|     {
|       member_id: "M1042",
|       rating: 4,
|       comment: "Good but the Digital edition had formatting issues.",
|       date: ISODate("2023-02-20")
|     }
|   ]
| }
| });
| {
|   acknowledged: true,
|   insertedId: ObjectId('6a95a0531b06dfe6c9d9a600')
| }
bookflowDB;> |
```

```
bookflowDB;> db.book_metadata.insertMany({
  title: "Ocean of Code",
  author: "A. Fernandes",
  genre: "Technology",
  publishedYear: 2021,
  formats: ["Digital", "Paperback"],
  specs: { pages: 210, fileSize_MB: 2.1, drm: false },
  avgRating: 4.2,
  reviews: [
    { member_id: "M2001", rating: 4, comment: "Solid Digital read, clean layout.", date: ISODate("2022-05-10") },
    { member_id: "M2002", rating: 5, comment: "Best programming book this year.", date: ISODate("2022-06-01") }
  ]
},
```

```
{
  title: "Whispers of Yesterday",
  author: "S. Kapoor",
  genre: "Fiction",
  publishedYear: 2019,
  formats: ["Hardcover", "Paperback"],
  specs: { pages: 410, binding: "Sewn" },
  avgRating: 3.8,
  reviews: [
    { member_id: "M3001", rating: 3, comment: "Slow start but strong ending.", date: ISODate("2020-01-15") },
    { member_id: "M3002", rating: 4, comment: "Beautifully written prose.", date: ISODate("2020-02-11") }
  ]
},
```

```
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a95a2721b06dfe6c9d9a601'),
    '1': ObjectId('6a95a2721b06dfe6c9d9a602'),
    '2': ObjectId('6a95a2721b06dfe6c9d9a603')
  }
}
bookflowDB;> |
```

2. Advanced Querying

a) Books rated above 4 (\$gt)

```
bookflowDB;> db.book_metadata.find(
  { avgRating: { $gt: 4 } },
  { title: 1, avgRating: 1, _id: 0 }
);
[
  { title: 'The Silent Algorithm', avgRating: 4.6 },
  { title: 'Ocean of Code', avgRating: 4.2 },
  { title: 'Quantum Horizons', avgRating: 4.9 }
]
bookflowDB;> |
```

b) Books available in specific formats (\$in)

```

bookflowDB;> db.book_metadata.find(
|   { formats: { $in: ["Digital", "Audiobook"] } },
|   { title: 1, formats: 1, _id: 0 }
| );
[
  {
    title: 'The Silent Algorithm',
    formats: [ 'Hardcover', 'Digital', 'Audiobook' ]
  },
  { title: 'Ocean of Code', formats: [ 'Digital', 'Paperback' ] },
  { title: 'Quantum Horizons', formats: [ 'Digital', 'Audiobook' ] }
]
bookflowDB;> |

```

c) Combine both with \$and (implicit)

```

    fileSize_MB: 4.2,
    drm: true,
    narrator: 'K. Rao',
    runtime_min: 610
  },
  avgRating: 4.6,
  reviews: [
    {
      member_id: 'M1001',
      rating: 5,
      comment: 'Excellent deep dive into digital systems design.',
      date: ISODate('2023-01-15T00:00:00.000Z')
    },
    {
      member_id: 'M1042',
      rating: 4,
      comment: 'Good but the Digital edition had formatting issues.',
      date: ISODate('2023-02-20T00:00:00.000Z')
    }
  ]
},
{
  _id: ObjectId('6a95a2721b06dfe6c9d9a601'),
  title: 'Ocean of Code',
  author: 'A. Fernandes',
  genre: 'Technology',
  publishedYear: 2021,
  formats: [ 'Digital', 'Paperback' ],
  specs: { pages: 210, fileSize_MB: 2.1, drm: false },
  avgRating: 4.2,

```

```

    avgRating: 4.2,
    reviews: [
      {
        member_id: 'M2001',
        rating: 4,
        comment: 'Solid Digital read, clean layout.',
        date: ISODate('2022-05-10T00:00:00.000Z')
      },
      {
        member_id: 'M2002',
        rating: 5,
        comment: 'Best programming book this year.',
        date: ISODate('2022-06-01T00:00:00.000Z')
      }
    ]
  },
  {
    _id: ObjectId('6a95a2721b06dfe6c9d9a603'),
    title: 'Quantum Horizons',
    author: 'R. Menon',
    genre: 'Science',
    publishedYear: 2023,
    formats: [ 'Digital', 'Audiobook' ],
    specs: { fileSize_MB: 5.6, drm: true, narrator: 'P. Iyer', runtime_min: 540 },
    avgRating: 4.9,
    reviews: [
      {
        member_id: 'M4001',
        rating: 5,
        comment: 'The Digital format made annotations so easy.'

```

```

      {
        _id: ObjectId('6a95a2721b06dfe6c9d9a603'),
        title: 'Quantum Horizons',
        author: 'R. Menon',
        genre: 'Science',
        publishedYear: 2023,
        formats: [ 'Digital', 'Audiobook' ],
        specs: { fileSize_MB: 5.6, drm: true, narrator: 'P. Iyer', runtime_min: 540 },
        avgRating: 4.9,
        reviews: [
          {
            member_id: 'M4001',
            rating: 5,
            comment: 'The Digital format made annotations so easy.',
            date: ISODate('2023-08-01T00:00:00.000Z')
          },
          {
            member_id: 'M4002',
            rating: 5,
            comment: 'Mind-bending and well narrated.',
            date: ISODate('2023-09-14T00:00:00.000Z')
          }
        ]
      }
    ]
  }
]
bookflowDB;> |

```

d) Regex query — books with "Digital" format

```
bookflowDB;> db.book_metadata.find(
|   { formats: { $regex: "Digital", $options: "i" } },
|   { title: 1, formats: 1 }
| );
[
  {
    _id: ObjectId('6a95a0531b06dfe6c9d9a600'),
    title: 'The Silent Algorithm',
    formats: [ 'Hardcover', 'Digital', 'Audiobook' ]
  },
  {
    _id: ObjectId('6a95a2721b06dfe6c9d9a601'),
    title: 'Ocean of Code',
    formats: [ 'Digital', 'Paperback' ]
  },
  {
    _id: ObjectId('6a95a2721b06dfe6c9d9a603'),
    title: 'Quantum Horizons',
    formats: [ 'Digital', 'Audiobook' ]
  }
]
```

e) Regex query — reviews mentioning "Digital" in comments (nested field)

```
bookflowDB;> db.book_metadata.find(
|   { "reviews.comment": { $regex: "Digital", $options: "i" } },
|   { title: 1, "reviews.$": 1 }
| );
[
  {
    _id: ObjectId('6a95a0531b06dfe6c9d9a600'),
    title: 'The Silent Algorithm',
    reviews: [
      {
        member_id: 'M1001',
        rating: 5,
        comment: 'Excellent deep dive into digital systems design.',
        date: ISODate('2023-01-15T00:00:00.000Z')
      }
    ]
  },
  {
    _id: ObjectId('6a95a2721b06dfe6c9d9a601'),
    title: 'Ocean of Code',
    reviews: [
      {
        member_id: 'M2001',
        rating: 4,
        comment: 'Solid Digital read, clean layout.',
        date: ISODate('2022-05-10T00:00:00.000Z')
      }
    ]
  }
]
```

```

    {
      _id: ObjectId('6a95a2721b06dfe6c9d9a601'),
      title: 'Ocean of Code',
      reviews: [
        {
          member_id: 'M2001',
          rating: 4,
          comment: 'Solid Digital read, clean layout.',
          date: ISODate('2022-05-10T00:00:00.000Z')
        }
      ]
    },
    {
      _id: ObjectId('6a95a2721b06dfe6c9d9a603'),
      title: 'Quantum Horizons',
      reviews: [
        {
          member_id: 'M4001',
          rating: 5,
          comment: 'The Digital format made annotations so easy.',
          date: ISODate('2023-08-01T00:00:00.000Z')
        }
      ]
    }
  ]
}

```

3. Aggregation Pipeline --"Top Rated" Report

Stage 1: filter books published after 2020

```
bookflowDB;> db.book_metadata.aggregate([
|   { $match: { publishedYear: { $gt: 2020 } } }
| ]);
|
| {
|   _id: ObjectId('6a95a0531b06dfe6c9d9a600'),
|   title: 'The Silent Algorithm',
|   author: 'R. Menon',
|   genre: 'Technology',
|   publishedYear: 2022,
|   formats: [ 'Hardcover', 'Digital', 'Audiobook' ],
|   specs: {
|     pages: 340,
|     fileSize_MB: 4.2,
|     drm: true,
|     narrator: 'K. Rao',
|     runtime_min: 610
|   },
|   avgRating: 4.6,
|   reviews: [
|     {
|       member_id: 'M1001',
|       rating: 5,
|       comment: 'Excellent deep dive into digital systems design.',
|       date: ISODate('2023-01-15T00:00:00.000Z')
|     },
|   ],
| }
```

```
    },
    {
      member_id: 'M1042',
      rating: 4,
      comment: 'Good but the Digital edition had formatting issues.',
      date: ISODate('2023-02-20T00:00:00.000Z')
    }
  ]
},
{
  _id: ObjectId('6a95a2721b06dfe6c9d9a601'),
  title: 'Ocean of Code',
  author: 'A. Fernandes',
  genre: 'Technology',
  publishedYear: 2021,
  formats: [ 'Digital', 'Paperback' ],
  specs: { pages: 210, fileSize_MB: 2.1, drm: false },
  avgRating: 4.2,
  reviews: [
    {
      member_id: 'M2001',
      rating: 4,
      comment: 'Solid Digital read, clean layout.',
      date: ISODate('2022-05-10T00:00:00.000Z')
    },
    {
      member_id: 'M2002',
      rating: 5,
      comment: 'Best programming book this year.',
      date: ISODate('2022-06-01T00:00:00.000Z')
    }
  ],
}
```

```

    date: ISODate('2022-06-01T00:00:00.000Z')
  }
]
},
{
  _id: ObjectId('6a95a2721b06dfe6c9d9a603'),
  title: 'Quantum Horizons',
  author: 'R. Menon',
  genre: 'Science',
  publishedYear: 2023,
  formats: [ 'Digital', 'Audiobook' ],
  specs: { fileSize_MB: 5.6, drm: true, narrator: 'P. Iyer', runtime_min: 540 },
  avgRating: 4.9,
  reviews: [
    {
      member_id: 'M4001',
      rating: 5,
      comment: 'The Digital format made annotations so easy.',
      date: ISODate('2023-08-01T00:00:00.000Z')
    },
    {
      member_id: 'M4002',
      rating: 5,
      comment: 'Mind-bending and well narrated.',
      date: ISODate('2023-09-14T00:00:00.000Z')
    }
  ]
}
]
}
]
bookflowDB> |

```

Stage 2: deconstruct the reviews array — one output doc per review

```

bookflowDB;> db.book_metadata.aggregate([
|   { $unwind: "$reviews" }
| ]);
|
| {
|   _id: ObjectId('6a95a0531b06dfe6c9d9a600'),
|   title: 'The Silent Algorithm',
|   author: 'R. Menon',
|   genre: 'Technology',
|   publishedYear: 2022,
|   formats: [ 'Hardcover', 'Digital', 'Audiobook' ],
|   specs: {
|     pages: 340,
|     fileSize_MB: 4.2,
|     drm: true,
|     narrator: 'K. Rao',
|     runtime_min: 610
|   },
|   avgRating: 4.6,
|   reviews: {
|     member_id: 'M1001',
|     rating: 5,
|     comment: 'Excellent deep dive into digital systems design.',
|     date: ISODate('2023-01-15T00:00:00.000Z')
|   }
| },
| }
|

```

```

{
  _id: ObjectId('6a95a0531b06dfe6c9d9a600'),
  title: 'The Silent Algorithm',
  author: 'R. Menon',
  genre: 'Technology',
  publishedYear: 2022,
  formats: [ 'Hardcover', 'Digital', 'Audiobook' ],
  specs: {
    pages: 340,
    fileSize_MB: 4.2,
    drm: true,
    narrator: 'K. Rao',
    runtime_min: 610
  },
  avgRating: 4.6,
  reviews: {
    member_id: 'M1042',
    rating: 4,
    comment: 'Good but the Digital edition had formatting issues.'
    date: ISODate('2023-02-20T00:00:00.000Z')
  }
},
{
  _id: ObjectId('6a95a2721b06dfe6c9d9a601'),
  title: 'Ocean of Code',
  author: 'A. Fernandes',
  genre: 'Technology',
  publishedYear: 2021,
  formats: [ 'Digital', 'Paperback' ],
  specs: { pages: 210, fileSize_MB: 2.1, drm: false },
}

```

```
{
  _id: ObjectId('6a95a0531b06dfe6c9d9a600'),
  title: 'The Silent Algorithm',
  author: 'R. Menon',
  genre: 'Technology',
  publishedYear: 2022,
  formats: [ 'Hardcover', 'Digital', 'Audiobook' ],
  specs: {
    pages: 340,
    fileSize_MB: 4.2,
    drm: true,
    narrator: 'K. Rao',
    runtime_min: 610
  },
  avgRating: 4.6,
  reviews: {
    member_id: 'M1042',
    rating: 4,
    comment: 'Good but the Digital edition had formatting issues.'
    date: ISODate('2023-02-20T00:00:00.000Z')
  }
},
{
  _id: ObjectId('6a95a2721b06dfe6c9d9a601'),
  title: 'Ocean of Code',
  author: 'A. Fernandes',
  genre: 'Technology',
  publishedYear: 2021,
  formats: [ 'Digital', 'Paperback' ],
  specs: { pages: 210, fileSize_MB: 2.1, drm: false },
}
```

```
{
  _id: ObjectId('6a95a0531b06dfe6c9d9a600'),
  title: 'The Silent Algorithm',
  author: 'R. Menon',
  genre: 'Technology',
  publishedYear: 2022,
  formats: [ 'Hardcover', 'Digital', 'Audiobook' ],
  specs: {
    pages: 340,
    fileSize_MB: 4.2,
    drm: true,
    narrator: 'K. Rao',
    runtime_min: 610
  },
  avgRating: 4.6,
  reviews: {
    member_id: 'M1042',
    rating: 4,
    comment: 'Good but the Digital edition had formatting issues.'
    date: ISODate('2023-02-20T00:00:00.000Z')
  }
},
{
  _id: ObjectId('6a95a2721b06dfe6c9d9a601'),
  title: 'Ocean of Code',
  author: 'A. Fernandes',
  genre: 'Technology',
  publishedYear: 2021,
  formats: [ 'Digital', 'Paperback' ],
  specs: { pages: 210, fileSize_MB: 2.1, drm: false },
}
```

```

{
  _id: ObjectId('6a95a0531b06dfe6c9d9a600'),
  title: 'The Silent Algorithm',
  author: 'R. Menon',
  genre: 'Technology',
  publishedYear: 2022,
  formats: [ 'Hardcover', 'Digital', 'Audiobook' ],
  specs: {
    pages: 340,
    fileSize_MB: 4.2,
    drm: true,
    narrator: 'K. Rao',
    runtime_min: 610
  },
  avgRating: 4.6,
  reviews: {
    member_id: 'M1042',
    rating: 4,
    comment: 'Good but the Digital edition had formatting issues.'
    date: ISODate('2023-02-20T00:00:00.000Z')
  }
},
{
  _id: ObjectId('6a95a2721b06dfe6c9d9a601'),
  title: 'Ocean of Code',
  author: 'A. Fernandes',
  genre: 'Technology',
  publishedYear: 2021,
  formats: [ 'Digital', 'Paperback' ],
  specs: { pages: 210, fileSize_MB: 2.1, drm: false },

```

Stage 3: group by title, compute average rating & review count

```

[
  {
    _id: 'Whispers of Yesterday',
    avgReviewRating: 3.5,
    totalReviews: 2,
    author: 'S. Kapoor'
  },
  {
    _id: 'The Silent Algorithm',
    avgReviewRating: 4.5,
    totalReviews: 2,
    author: 'R. Menon'
  },
  {
    _id: 'Ocean of Code',
    avgReviewRating: 4.5,
    totalReviews: 2,
    author: 'A. Fernandes'
  },
  {
    _id: 'Quantum Horizons',
    avgReviewRating: 5,
    totalReviews: 2,
    author: 'R. Menon'
  }
]
bookflowDB;> |

```

Stage 4: sort descending by average rating

```
bookFlowDB;> db.book_metadata.aggregate([
| { $sort: { avgReviewRating: -1 } },
| ])
[
| {
|   _id: ObjectId('6a95a0531b06dfe6c9d9a600'),
|   title: 'The Silent Algorithm',
|   author: 'R. Menon',
|   genre: 'Technology',
|   publishedYear: 2022,
|   formats: [ 'Hardcover', 'Digital', 'Audiobook' ],
|   specs: {
|     pages: 340,
|     fileSize_MB: 4.2,
|     drm: true,
|     narrator: 'K. Rao',
|     runtime_min: 610
|   },
|   avgRating: 4.6,
|   reviews: [
|     {
|       member_id: 'M1001',
|       rating: 5,
|       comment: 'Excellent deep dive into digital systems design.',
|       date: ISODate('2023-01-15T00:00:00.000Z')
|     },
|     {
|       member_id: 'M1042',
|       rating: 4,
|       comment: 'Good but the Digital edition had formatting issues.'
|       date: ISODate('2023-02-20T00:00:00.000Z')
|     }
|   ]
| }
| ]
```

```
{
|   _id: ObjectId('6a95a2721b06dfe6c9d9a601'),
|   title: 'Ocean of Code',
|   author: 'A. Fernandes',
|   genre: 'Technology',
|   publishedYear: 2021,
|   formats: [ 'Digital', 'Paperback' ],
|   specs: { pages: 210, fileSize_MB: 2.1, drm: false },
|   avgRating: 4.2,
|   reviews: [
|     {
|       member_id: 'M2001',
|       rating: 4,
|       comment: 'Solid Digital read, clean layout.',
|       date: ISODate('2022-05-10T00:00:00.000Z')
|     },
|     {
|       member_id: 'M2002',
|       rating: 5,
|       comment: 'Best programming book this year.',
|       date: ISODate('2022-06-01T00:00:00.000Z')
|     }
|   ]
| },
| {
|   _id: ObjectId('6a95a2721b06dfe6c9d9a602'),
|   title: 'Whispers of Yesterday',
|   author: 'S. Kapoor',
|   genre: 'Fiction',
|   publishedYear: 2019,
|   formats: [ 'Hardcover', 'Paperback' ],
|   specs: { pages: 410, binding: 'Sewn' },
|   avgRating: 3.8,
|   reviews: [
|     {
|
```

```

    {
      member_id: 'M3001',
      rating: 3,
      comment: 'Slow start but strong ending.',
      date: ISODate('2020-01-15T00:00:00.000Z')
    },
    {
      member_id: 'M3002',
      rating: 4,
      comment: 'Beautifully written prose.',
      date: ISODate('2020-02-11T00:00:00.000Z')
    }
  ]
},
{
  _id: ObjectId('6a95a2721b06dfe6c9d9a603'),
  title: 'Quantum Horizons',
  author: 'R. Menon',
  genre: 'Science',
  publishedYear: 2023,
  formats: [ 'Digital', 'Audiobook' ],
  specs: { fileSize_MB: 5.6, drm: true, narrator: 'P. Iyer', runtime_min: 540 },
  avgRating: 4.9,
  reviews: [
    {
      member_id: 'M4001',
      rating: 5,
      comment: 'The Digital format made annotations so easy.',
      date: ISODate('2023-08-01T00:00:00.000Z')
    },
    {
      member_id: 'M4002',
      rating: 5,
      comment: 'Mind-bending and well narrated.',
      date: ISODate('2023-09-14T00:00:00.000Z')
    }
  ]
}
]
}

```

```

  _id: ObjectId('6a95a2721b06dfe6c9d9a603'),
  title: 'Quantum Horizons',
  author: 'R. Menon',
  genre: 'Science',
  publishedYear: 2023,
  formats: [ 'Digital', 'Audiobook' ],
  specs: { fileSize_MB: 5.6, drm: true, narrator: 'P. Iyer', runtime_min: 540 },
  avgRating: 4.9,
  reviews: [
    {
      member_id: 'M4001',
      rating: 5,
      comment: 'The Digital format made annotations so easy.',
      date: ISODate('2023-08-01T00:00:00.000Z')
    },
    {
      member_id: 'M4002',
      rating: 5,
      comment: 'Mind-bending and well narrated.',
      date: ISODate('2023-09-14T00:00:00.000Z')
    }
  ]
}
]
}
bookFlowDB>

```

Stage 5: reshape output

```
[
  {
    author: 'R. Menon',
    title: ObjectId('6a95a0531b06dfe6c9d9a600'),
    avgReviewRating: null
  },
  {
    author: 'A. Fernandes',
    title: ObjectId('6a95a2721b06dfe6c9d9a601'),
    avgReviewRating: null
  },
  {
    author: 'S. Kapoor',
    title: ObjectId('6a95a2721b06dfe6c9d9a602'),
    avgReviewRating: null
  },
  {
    author: 'R. Menon',
    title: ObjectId('6a95a2721b06dfe6c9d9a603'),
    avgReviewRating: null
  }
]
bookflowDB;> |
```

4.Indexing for Speed

a) Create a text index on review comments

```
bookflowDB;> db.book_metadata.createIndex({ "reviews.comment": "text" });
reviews.comment_text
bookflowDB;> |
```

b) Query using the text index

```
bookflowDB;> db.book_metadata.find(
|   { $text: { $search: "Digital" } },
|   { title: 1, score: { $meta: "textScore" } }
| ).sort({ score: { $meta: "textScore" } });
[
  {
    _id: ObjectId('6a95a0531b06dfe6c9d9a600'),
    title: 'The Silent Algorithm',
    score: 1.1833333333333333
  },
  {
    _id: ObjectId('6a95a2721b06dfe6c9d9a603'),
    title: 'Quantum Horizons',
    score: 0.6
  },
  {
    _id: ObjectId('6a95a2721b06dfe6c9d9a601'),
    title: 'Ocean of Code',
    score: 0.6
  }
]
bookflowDB;> |
```

c) Verify index usage with .explain()

```
]
bookflowDB;> db.book_metadata.find(
|   { $text: { $search: "Digital" } }
| ).explain("executionStats");
{
  explainVersion: '1',
  queryPlanner: {
    namespace: 'bookflowDB.book_metadata',
    parsedQuery: {
      '$text': {
        '$search': 'Digital',
        '$language': 'english',
        '$caseSensitive': false,
        '$diacriticSensitive': false
      }
    },
    indexFilterSet: false,
    queryHash: '4FB85230',
    planCacheShapeHash: '4FB85230',
    planCacheKey: 'F34F3876',
    optimizationTimeMillis: 0,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExplodeReached: false,
    prunedSimilarIndexes: false,
    winningPlan: {
      isCached: false,
      stage: 'TEXT_MATCH',
      nss: 'bookflowDB.book_metadata',
      indexPrefix: {},
      indexName: 'reviews.comment_text',
      parsedTextQuery: {
        terms: [ 'digit' ],
        negatedTerms: [],
        phrases: [],
        negatedPhrases: []
      },
      textIndexVersion: 3,
    }
  }
}
```

```
textIndexVersion: 3,
inputStage: {
  stage: 'FETCH',
  nss: 'bookFlowDB;.book_metadata',
  inputStage: {
    stage: 'IXSCAN',
    nss: 'bookFlowDB;.book_metadata',
    keyPattern: { _fts: 'text', _ftsx: 1 },
    indexName: 'reviews.comment_text',
    isMultiKey: true,
    isUnique: false,
    isSparse: false,
    isPartial: false,
    indexVersion: 2,
    direction: 'backward',
    indexBounds: {}
  }
},
rejectedPlans: []
},
executionStats: {
  executionSuccess: true,
  nReturned: 3,
  executionTimeMillis: 7,
  totalKeysExamined: 3,
  totalDocsExamined: 3,
  executionStages: {
    isCached: false,
    stage: 'TEXT_MATCH',
    nReturned: 3,
    executionTimeMillisEstimate: 0,
    works: 4,
    advanced: 3,
    needTime: 0,
    needYield: 0,
    saveState: 0,
    restoreState: 0,
    isEOF: 1,
    nss: 'bookFlowDB;.book_metadata',
  }
}
```

```
isEOF: 1,
nss: 'bookflowDB;.book_metadata',
indexPrefix: {},
indexName: 'reviews.comment_text',
parsedTextQuery: {
  terms: [ 'digit' ],
  negatedTerms: [],
  phrases: [],
  negatedPhrases: []
},
textIndexVersion: 3,
docsRejected: 0,
inputStage: {
  stage: 'FETCH',
  nReturned: 3,
  executionTimeMillisEstimate: 0,
  works: 4,
  advanced: 3,
  needTime: 0,
  needYield: 0,
  saveState: 0,
  restoreState: 0,
  isEOF: 1,
  nss: 'bookflowDB;.book_metadata',
  docsExamined: 3,
  alreadyHasObj: 0,
  inputStage: {
    stage: 'IXSCAN',
    nReturned: 3,
    executionTimeMillisEstimate: 0,
    works: 4,
    advanced: 3,
    needTime: 0,
    needYield: 0,
    saveState: 0,
    restoreState: 0,
    isEOF: 1,
    nss: 'bookflowDB;.book_metadata',
    keyPattern: { _fts: 'text', _ftsx: 1 },
    indexName: 'reviews.comment_text',
```

```
nReturned: 3,
executionTimeMillisEstimate: 0,
works: 4,
advanced: 3,
needTime: 0,
needYield: 0,
saveState: 0,
restoreState: 0,
isEOF: 1,
nss: 'bookflowDB;.book_metadata',
docsExamined: 3,
alreadyHasObj: 0,
inputStage: {
  stage: 'IXSCAN',
  nReturned: 3,
  executionTimeMillisEstimate: 0,
  works: 4,
  advanced: 3,
  needTime: 0,
  needYield: 0,
  saveState: 0,
  restoreState: 0,
  isEOF: 1,
  nss: 'bookflowDB;.book_metadata',
  keyPattern: { _fts: 'text', _ftsx: 1 },
  indexName: 'reviews.comment_text',
  isMultiKey: true,
  isUnique: false,
  isSparse: false,
  isPartial: false,
  indexVersion: 2,
  direction: 'backward',
  indexBounds: {},
  keysExamined: 3,
  seeks: 1,
  dupsTested: 3,
  dupsDropped: 0,
  peakTrackedMemBytes: 26
}
```

```

        peakTrackedMemBytes: 26
      }
    }
  },
  queryShapeHash: '8FFE3937A76E5AF10E5F9D148EF8685D5A45B1C52D0EFBA5D027172EBB8FA3D7',
  peakTrackedMemBytes: Long('26'),
  command: {
    find: 'book_metadata',
    filter: { '$text': { '$search': 'Digital' } },
    '$db': 'bookflowDB;'
  },
  serverInfo: {
    host: 'DESKTOP-P0IK742',
    port: 27017,
    version: '8.3.8',
    gitVersion: '35e8c8a57f78157ed9fac1a9e90ee6c1818adab6'
  },
  serverParameters: {
    internalQueryFacetBufferSizeBytes: 104857600,
    internalDocumentSourceGroupMaxMemoryBytes: 104857600,
    internalQueryMaxBlockingSortMemoryUsageBytes: 104857600,
    internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
    internalQueryFacetMaxOutputDocSizeBytes: 104857600,
    internalLookupStageIntermediateDocumentMaxSizeBytes: 104857600,
    internalQueryProhibitBlockingMergeOnMongoS: 0,
    internalQueryMaxAddToSetBytes: 104857600,
    internalQueryFrameworkControl: 'trySbeRestricted',
    internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
  },
  ok: 1
}
bookflowDB;> |

```

d) Compare against a non-indexed regex scan

```

bookflowDB;> db.book_metadata.find(
| { "reviews.comment": { $regex: "Digital", $options: "i" } }
| ).explain("executionStats");
{
  explainVersion: '1',
  queryPlanner: {
    namespace: 'bookflowDB;book_metadata',
    parsedQuery: { 'reviews.comment': { '$regex': 'Digital', '$options': 'i' } }
    indexFilterSet: false,
    queryHash: '819E6067',
    planCacheShapeHash: '819E6067',
    planCacheKey: '29FFD58A',
    optimizationTimeMillis: 0,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExplodeReached: false,
    prunedSimilarIndexes: false,
    winningPlan: {
      isCached: false,
      stage: 'COLLSCAN',
      filter: { 'reviews.comment': { '$regex': 'Digital', '$options': 'i' } },
      nss: 'bookflowDB;book_metadata',
      direction: 'forward'
    },
    rejectedPlans: []
  },
  executionStats: {
    executionSuccess: true,
    nReturned: 3,
    executionTimeMillis: 0,
    totalKeysExamined: 0,
    totalDocsExamined: 4,
    executionStages: {
      isCached: false,
      stage: 'COLLSCAN',
      filter: { 'reviews.comment': { '$regex': 'Digital', '$options': 'i' } },
      nReturned: 3,
      executionTimeMillisEstimated: 0
    }
  }
}

```

```

    executionTimeMillisEstimate: 0,
    works: 5,
    advanced: 3,
    needTime: 1,
    needYield: 0,
    saveState: 0,
    restoreState: 0,
    isEOF: 1,
    nss: 'bookflowDB;.book_metadata',
    direction: 'forward',
    docsExamined: 4
  }
},
queryShapeHash: '5A944951605A37CD22FC91D534B941C78F0CA6F511E9E9CCD598D758D0E8C55F',
command: {
  find: 'book_metadata',
  filter: { 'reviews.comment': { '$regex': 'Digital', '$options': 'i' } },
  '$db': 'bookflowDB;'
},
serverInfo: {
  host: 'DESKTOP-P0IK742',
  port: 27017,
  version: '8.3.8',
  gitVersion: '35e8c8a57f78157ed9fac1a9e90ee6c1818adab6'
},
serverParameters: {
  internalQueryFacetBufferSizeBytes: 104857600,
  internalDocumentSourceGroupMaxMemoryBytes: 104857600,
  internalQueryMaxBlockingSortMemoryUsageBytes: 104857600,
  internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
  internalQueryFacetMaxOutputDocSizeBytes: 104857600,
  internalLookupStageIntermediateDocumentMaxSizeBytes: 104857600,
  internalQueryProhibitBlockingMergeOnMongoS: 0,
  internalQueryMaxAddToSetBytes: 104857600,
  internalQueryFrameworkControl: 'trySbeRestricted',
  internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
},
ok: 1
}
bookflowDB;> |

```

e) Supporting index for common queries (multikey index)

```

bookflowDB;> db.book_metadata.createIndex({ formats: 1 });
formats_1
bookflowDB;> db.book_metadata.createIndex({ avgRating: -1 });
avgRating_-1

```

```

avgRating: -1
bookflowDB:> db.book_metadata.find({ avgRating: { $gt: 4 } }).explain("executionStats");
{
  explainVersion: '1',
  queryPlanner: {
    namespace: 'bookflowDB:book_metadata',
    parsedQuery: { avgRating: { '$gt': 4 } },
    indexFilterSet: false,
    queryHash: 'E21FCB34',
    planCacheShapeHash: 'E21FCB34',
    planCacheKey: '6A44442CC',
    optimizationTimeMillis: 1,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExplodeReached: false,
    prunedSimilarIndexes: false,
    winningPlan: {
      isCached: false,
      stage: 'FETCH',
      nss: 'bookflowDB:book_metadata',
      inputStage: {
        stage: 'IXSCAN',
        nss: 'bookflowDB:book_metadata',
        keyPattern: { avgRating: -1 },
        indexName: 'avgRating_-1',
        isMultiKey: false,
        multiKeyPaths: { avgRating: [] },
        isUnique: false,
        isSparse: false,
        isPartial: false,
        indexVersion: 2,
        direction: 'forward',
        indexBounds: { avgRating: [ '[inf, 4)' ] }
      }
    },
    rejectedPlans: []
  },
  executionStats: {
    executionSuccess: true,
    nReturned: 3,
    executionTimeMillis: 13,
    totalKeysExamined: 3,
    totalDocsExamined: 3,
    executionStages: {
      isCached: false,

```

```

    },
    executionStats: {
      executionSuccess: true,
      nReturned: 3,
      executionTimeMillis: 13,
      totalKeysExamined: 3,
      totalDocsExamined: 3,
      executionStages: {
        isCached: false,
        stage: 'FETCH',
        nReturned: 3,
        executionTimeMillisEstimate: 10,
        works: 4,
        advanced: 3,
        needTime: 0,
        needYield: 0,
        saveState: 0,
        restoreState: 0,
        isEOF: 1,
        nss: 'bookflowDB:book_metadata',
        docsExamined: 3,
        alreadyHasObj: 0,
        inputStage: {
          stage: 'IXSCAN',
          nReturned: 3,
          executionTimeMillisEstimate: 10,
          works: 4,
          advanced: 3,
          needTime: 0,
          needYield: 0,
          saveState: 0,
          restoreState: 0,
          isEOF: 1,
          nss: 'bookflowDB:book_metadata',
          keyPattern: { avgRating: -1 },
          indexName: 'avgRating_-1',
          isMultiKey: false,
          multiKeyPaths: { avgRating: [] },
          isUnique: false,
          isSparse: false,
          isPartial: false,
          indexVersion: 2,
          direction: 'forward',
          indexBounds: { avgRating: [ '[inf, 4)' ] },
          keysExamined: 3,
          seeks: 1,

```

```
    indexBounds: { avgRating: [ '[inf, 4)' ] },
    keysExamined: 3,
    seeks: 1,
    dupsTested: 0,
    dupsDropped: 0,
    peakTrackedMemBytes: 0
  }
},
queryShapeHash: '4E4E4126D4A6F49EEACDF870A223F71E13012D98E4BF99F3F69BB2473486F1C0',
command: {
  find: 'book_metadata',
  filter: { avgRating: { '$gt': 4 } },
  '$db': 'bookflowDB;'
},
serverInfo: {
  host: 'DESKTOP-P0IK742',
  port: 27017,
  version: '8.3.8',
  gitVersion: '35e8c8a57f78157ed9fac1a9e90ee6c1818adab6'
},
serverParameters: {
  internalQueryFacetBufferSizeBytes: 104857600,
  internalDocumentSourceGroupMaxMemoryBytes: 104857600,
  internalQueryMaxBlockingSortMemoryUsageBytes: 104857600,
  internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
  internalQueryFacetMaxOutputDocSizeBytes: 104857600,
  internalLookupStageIntermediateDocumentMaxSizeBytes: 104857600,
  internalQueryProhibitBlockingMergeOnMongoS: 0,
  internalQueryMaxAddToSetBytes: 104857600,
  internalQueryFrameworkControl: 'trySbeRestricted',
  internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
},
ok: 1
}
bookflowDB->
```